PCB Design Review Checklist

PCB Part Number and Revision:

_____

Assembled PCB P/N and Revision:

_____

Engineer:

_____

Date:

_____

Attendees:

_____

_____

Part Placement

1. Polarized components checked
2. Check the orientation of all connectors
3. Minimum component body spacing
4. Bypass capacitors located close to IC power pins
5. Verify that all series terminators are located near the source
6. I/O drivers near where their signals leave the board
7. PCB has ground turrets, power rail test points, and test points for important signals, all labeled
8. EMI and RFI filtering as close as possible to exit and entry points in shielded areas
9. Mounting holes electrically isolated or not
10. Proper mounting hole clearance for hardware
11. SMD pad shapes checked
12. Sufficient clearance for socketed ICs
13. SMD component orientation consistent
14. Footprints verified against manufacturer datasheet (new parts)

Routing

15. Check for traces running under noisy or sensitive components
16. Provide multiple vias for high current and/or low impedance traces
17. Component and trace keepout areas observed
18. Ground planes where appropriate
19. Differential lines routed appropriately

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Dimensions

20. Hole diameter on drawing are finished sizes, after plating.
21. Silkscreen legend text weight >=6 mils
22. Silkscreen legend text height >=30 mils
23. Pads >=5 mils larger than finished hole sizes
24. Test pads 200 mils from edge of board
25. Traces >= 15 mils from edge of PCB
26. Trace width sufficient for current carried
27. Sufficient clearance for high voltage traces

Silk Screen

28. No silkscreen text over vias (if vias not soldermasked) or holes
29. All legend text reads in one or two directions
30. Company logo / name in silkscreen legend
31. Date code on PCB
32. PCB part number on PCB
33. Assembly part number on PCB
34. PCB revision on silkscreen legend
35. Assembly revision blank on silkscreen legend
36. Serial number blank on silkscreen legend
37. All silkscreen text located to be readable when the board is populated
38. All ICs have pin one clearly marked, visible even when chip is installed

Other

39. DRC and Electrical Test Error Clean
40. High frequency circuitry precautions observed
41. Check hole diameters for odd components: rectangular pins, spring pins
42. Soldermask in relation to via coverage (tenting)
43. Soldermask swell checked
44. Fiducials allow for triangulation (multiple sites for flex)
45. PCB thickness, material, copper weight noted
46. Thermal reliefs for internal power layers
47. PCB layout panelized correctly

Documentation and Release Package

Altium Generated PDF:

A Schematic

Composite drawing:

B One page per metal layer

C One page for Paste, Mask, and Silk (per side) – Paste overlays mask

D Mechanical and Drills

E Used layers only

Items on layout:

A Drill Chart

B Layer Guide

C Title Block

Title Block Includes: N/A

_____A Title

_____B Description

_____C PCB Part Number

_____D Revision History

_____E Company

_____F Address

_____G Confidentiality Statement

_____H Engineer

Readme:

A Company Name

B Date

C PCB Part Number

D Description

E Revision

F Contact Information

G Stack Drawing

H Plating/Finish

I RoHS Compliance Statement

J UL Designation

K Conformance Statement for 6012/6013 and Class

L Material Construction (FR4 – etc)
M Any impedance Requirements
N List of all layer filenames and Descriptions
O Stiffener / Flex Circuit Information Explicit

PCB Package Requirements (.zip):

A Correct File name
B Schematic
C Fabrication Drawing
D Gerber Files
E OSB Files
F NC Drill Files
G Assembly drawing
H BOM
I Pick and place files
J README / Fabrication Document
K 3D Render of PCB
L STEP File